

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 - ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO2 - Manipulation (BL3, BL4)	Assessment:	Assessment Tool 1 - Analytical Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	Niruthika Sri Prakash	Reg. No.:	192511294
Section / Batch:	2 nd year / CSE	Date:	20/07/2026

Code of Conduct

I, Niruthika Sri Prakash (192511294) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Niruthika Sri Prakash

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
GRAND TOTAL:		100 Marks	

CO 2 - Assessment Tool 1

Analytical Problem Solving

Name : Nikhitha Sri Prakash

Reg no: 192511294

Course Code : CSA0305

Course : Data Structures

- ① Perform the following operations in LIFO data structures having an array size of 5. Push(45), Push(22), Push(11), Push(77), Pop(), Push(44), Push(33), Pop(), Pop(), Pop(). Print the peak element of data structure and its index position. With a neat sketch explained in detail.

Given operations :

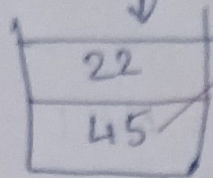
Push(45), Push(22), Push(11), Push(77), Pop()
Push(44), Push(33), Pop(), Pop(), Pop().

operation	Stack contents (Bottom $\rightarrow$ Top)	Top Index
Initial	Empty	-1
Push(45)	45	0
Push(22)	45, 22	1
Push(11)	45, 22, 11	2
Push(77)	45, 22, 11, 77	3
Pop()	45, 22, 11	2
Push(44)	45, 22, 11, 44	3
Push(33)	45, 22, 11, 44, 33	4
Pop()	45, 22, 11, 44	3
Pop()	45, 22, 11	2
Pop()	45, 22	1

Peak Top element = 22

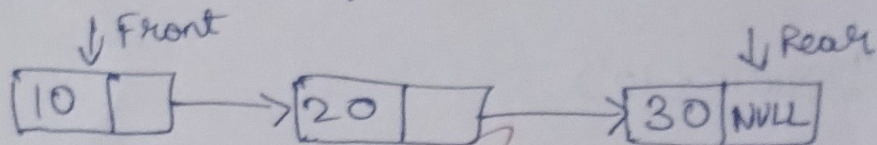
Index Position = 1

Final Stack TOP



- ② How do you implement Queue data Structure using Single Linked List? With a neat diagram discussing Enqueue and Dequeue operations using SLL.

Queue Structure :



(i) Enqueue(40)

creat a new node.

Insert x into the node.

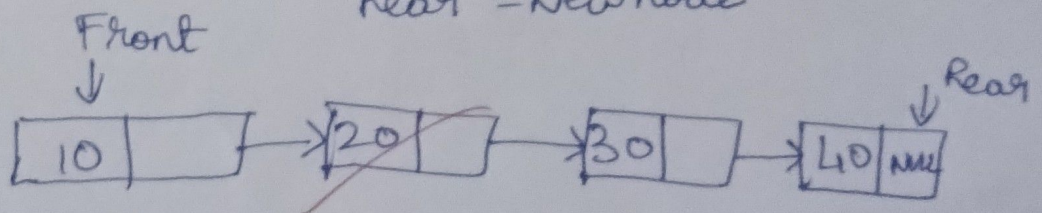
IF front == NULL

front = Rear = New node

ELSE

Rear -> next = New node

Rear = New node



(ii) Dequeue()

If Front == NULL

Queue Underflow

Temp = Front

Front = Front → next

Delete Temp

If Front == NULL

Rear = NULL

Dequeue(10)

